

Programme: B E – Information Science & Engineering
Internal Assessment – I

TERM : 10-Sep-2025 to 31-Dec-2025	COURSE NAME: Computer Networks
DATE : 30-10-202 _ TIME: 11AM to 12PM	COURSE CODE : IS53
MAX MARKS: 30	PORTIONS : L1 – L18



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	We have a channel with a 2-Mhz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level? (to calculate signal level consider the bit rate to be 4Mbps less than the upper limit).	L3	1	4
b	Differentiate between connection oriented and connectionless services.	L2	1	4
c	Define cyclic codes and linear block codes. Given the dataword 101011 and the divisor 10011, generate the codeword using CRC polynomial division.	L3	2	7
2.a	Compare mesh topology with star topology. Draw a hybrid topology with a star backbone and three bus networks.	L4	1	4
b	Convert the digital data 11001010 to digital signal by applying AMI and differential Manchester line coding schemes.	L3	1	4
c	With a neat diagram, discuss the different fields of ARP packet.	L2	2	7
3.a	Compare bit stuffing and byte stuffing with suitable examples.	L4	2	4
b	Calculate the Internet check sum for the text "ISE DEPARTMENT" at sender and receiver side (assume the following hexadecimal ASCII values: A=41, D=44, E=45, I=49, M=4D, N=4E, P=50, R=52, S=53, T=54, space=32)	L3	2	4
c	What is transmission impairment? Briefly explain its causes with necessary diagrams.	L2	1	7

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create



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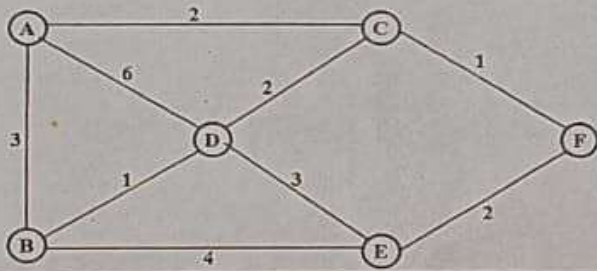
TERM : 10-Sep-2025 to 31-Dec-2025	COURSE NAME: Computer Networks
DATE : 22-12-2025 TIME: 11AM to 12PM	COURSE CODE : IS53
MAX MARKS: 30	PORTIONS : L25 – L56



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Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Elaborate on the different error reporting messages and query messages supported by ICMP.	L2	3	6
b	Consider $m=2$, and with necessary diagrams justify the statement: "Size of the sender window must be equal to 2^{m-1} in Selective Repeat ARQ Protocol"	L4	4	6
c	Explain the three range of port numbers defined by ICANN.	L2	5	3
2.a	With a neat diagram, discuss the procedure for NAT address translation using one IP address.	L2	3	6
b	Briefly discuss the working of stop and wait protocol for noisy channel with necessary diagrams.	L2	4	6
c	With respect to the congestion detection phase, highlight the behavior of Reno TCP.	L2	5	3
3.a	Applying Link State Routing, calculate the shortest path from the root node A to all other nodes in the following network configuration. 	L3	3	6
b	Discuss the different strategies to achieve transition from IPv4 to IPv6	L2	4	6
c	Explain how recursive resolution solves the name-address conflict in DNS.	L2	5	3

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create